

Sensitivity and Elasticity Analysis

In this assignment, we assume that the Leslie matrix has already been constructed. We will investigate how changes in individual fecundity and survival parameters affect the dominant eigenvalue and, consequently, the long-term population growth rate.

Our modeling questions are:

Which demographic parameters have the greatest influence on long-term population growth?

If one entry of the Leslie matrix changes slightly, how much does the dominant eigenvalue change?

Learning Objectives

After completing this assignment, you should be able to

- compute the left and right eigenvectors associated with the dominant eigenvalue of a Leslie matrix;
- construct sensitivity and elasticity matrices;
- visualize sensitivity and elasticity matrices using annotated heatmaps;
- interpret the biological meaning of sensitivity and elasticity analyses.

Background

Let $L = (a_{ij})$ be a Leslie matrix with a simple dominant eigenvalue λ . Let $\mathbf{w}$ and $\mathbf{v}$ be the corresponding right and left eigenvectors:

$$L\mathbf{w} = \lambda\mathbf{w}, \quad \mathbf{v}^T L = \lambda\mathbf{v}^T.$$

The right eigenvector $\mathbf{w}$, when normalized so that its entries sum to one, gives the stable age distribution. The left eigenvector $\mathbf{v}$ gives the reproductive values of the age classes.

Sensitivity

The sensitivity of λ to a matrix entry a_{ij} is

$$s_{ij} = \frac{\partial \lambda}{\partial a_{ij}}.$$

It measures the approximate absolute change in the dominant eigenvalue produced by a small absolute change in a_{ij} . For a simple dominant eigenvalue, first-order perturbation theory gives

$$s_{ij} = \frac{v_i w_j}{\mathbf{v}^T \mathbf{w}}. \quad (1)$$

If the eigenvectors are normalized so that $\mathbf{v}^T \mathbf{w} = 1$, then $s_{ij} = v_i w_j$. For a small perturbation Δa_{ij} ,

$$\Delta \lambda \approx s_{ij} \Delta a_{ij}.$$

Elasticity

Sensitivity measures absolute effects. Because Leslie-matrix entries may differ in magnitude and units, proportional effects are often more useful for comparison.

The elasticity of λ with respect to a_{ij} is

$$e_{ij} = \frac{a_{ij}}{\lambda} s_{ij}. \quad (2)$$

Equivalently,

$$e_{ij} = \frac{\partial \log \lambda}{\partial \log a_{ij}}.$$

Thus, e_{ij} measures the proportional change in λ resulting from a small proportional change in a_{ij} . For example, an elasticity of 0.20 means that a 1% increase in that matrix entry is associated, locally, with an approximate 0.20% increase in λ .

1 Enter the Leslie Matrix

Use the given Leslie matrix.

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.linalg import eig

L = np.array([
    [0, 0.036, 0.204, 0.255, 0.144, 0.118],
    [0.789, 0, 0, 0, 0, 0],
    [0, 0.860, 0, 0, 0, 0],
    [0, 0, 0.971, 0, 0, 0],
    [0, 0, 0, 0.523, 0, 0],
    [0, 0, 0, 0, 0.750, 0]
])

print(L)
print("Shape:", L.shape)
```

Questions

1. What do the entries in the first row of the Leslie matrix represent? Why are they located in the first row?

2. What do the entries on the subdiagonal represent? Explain why they appear only on the subdiagonal.
3. Why are all other entries of the Leslie matrix equal to zero?

2 Compute the Dominant Eigenvalue and Right Eigenvector

Use `eig` to compute the eigenvalues and right eigenvectors.

```
eigvals, right_vecs = eig(L)

# TODO:
# Find the index of the eigenvalue having the largest magnitude.
dominant_index =

# TODO:
# Extract the dominant eigenvalue and retain its real part.
lambda =

# TODO:
# Extract the corresponding right eigenvector.
w =

# Reverse the sign if needed.
if np.sum(w) < 0:
    w = -w

# TODO:
# Normalize w so its entries sum to one.
w =

print("Dominant eigenvalue:", lambda)
print("Normalized right eigenvector:", w)
```

Checks

Verify numerically that

$$Lw \approx \lambda w$$

using

```
print(np.allclose(L @ w, lambda * w))
```

3 Compute the Left Eigenvector

A left eigenvector of L is a right eigenvector of L^T .

```
eigvals_left, left_vecs = eig(L.T)

# The eigenvalues returned for L and L.T may not appear
# in the same order. Find the eigenvalue of L.T closest
```

```

# to lambda.
left_index = np.argmin(
    np.abs(eigvals_left - lambda)
)

# TODO:
# Extract the corresponding left eigenvector
# and retain its real part.
v =

# Reverse the sign if needed.
if np.sum(v) < 0:
    v = -v

# TODO:
# Normalize v so that v @ w = 1.
v =

print("Normalized left eigenvector:", v)
print("v dot w =", v @ w)

```

Verify numerically that

$$\mathbf{v}^T L \approx \lambda \mathbf{v}^T.$$

```

print(np.allclose(v @ L, lambda * v))

```

Questions

1. Why do we compute eigenvectors of L^T to obtain the left eigenvector?
2. Why is the normalization $\mathbf{v}^T \mathbf{w} = 1$ convenient?
3. What biological interpretation is assigned to the entries of $\mathbf{v}$?

4 Construct the Sensitivity Matrix

Equation (1) can be computed using an outer product.

```

# TODO:
# Construct the sensitivity matrix.
# Because v @ w = 1, use the outer product of v and w.

sensitivity =

print("Sensitivity matrix:")
print(np.round(sensitivity, 3))

```

Check the dimensions:

```

print(sensitivity.shape)

```

Interpretation

For a selected biologically meaningful entry a_{ij} , use the approximation

$$\Delta\lambda \approx s_{ij}\Delta a_{ij}$$

to estimate the effect of a small absolute increase in that entry.

1. Which fecundity entry has the largest sensitivity?
2. Which survival probability has the largest sensitivity?

5 Construct the Elasticity Matrix

Use Equation (2).

```
# TODO:  
# Compute the elasticity matrix entry by entry  
# using array operations.  
  
elasticity =  
  
print("Elasticity matrix:")  
print(np.round(elasticity, 3))
```

For a projection matrix with a simple dominant eigenvalue, the elasticities of all entries sum to approximately one.

```
print("Sum of elasticities:", elasticity.sum())
```

Questions

1. Which fecundity entry has the largest elasticity?
2. Which survival probability has the largest elasticity?
3. Are the entries with the largest sensitivity and elasticity the same?
4. Why can sensitivity and elasticity produce different rankings?

6 Select Biologically Meaningful Entries and Create Heatmaps

For this Leslie matrix, the biologically meaningful nonzero entries are

- fecundities in the first row;
- survival probabilities on the subdiagonal.

Create a mask that hides all other cells.

```

mask = np.ones_like(L, dtype=bool)

# Fecundities  $F_2$  through  $F_{6+}$ 
mask[0, 1:6] = False

# Survival entries  $S_1$  through  $S_5$ 
for i in range(1, 6):
    mask[i, i - 1] = False

```

Explain why the entry F_1 is excluded and why the terminal survival entry is not displayed.

Create Heatmaps

Use the following age labels.

```

ages = [
    "Age-1", "Age-2", "Age-3",
    "Age-4", "Age-5", "Age-6+"
]

```

Sensitivity heatmap

Complete the code.

```

plt.figure(figsize=(8, 6))

sns.heatmap(
    sensitivity,
    mask=mask,
    annot=True,
    fmt=".3f",
    cmap="YlGnBu",
    xticklabels=ages,
    yticklabels=ages,
    cbar_kws={"label": "Sensitivity"},
    annot_kws={"size": 14}
)

plt.xlabel("To age / fecundity", fontsize=16)
plt.ylabel("From age", fontsize=16)
plt.xticks(fontsize=14)
plt.yticks(fontsize=14)
plt.tight_layout()

plt.savefig(
    "sensitivity_matrix.pdf",
    bbox_inches="tight"
)

plt.show()

```

Elasticity heatmap

Create a second heatmap using the same structure, but replace the data and color-bar label appropriately.

```
# TODO:  
# Create and save the elasticity heatmap  
# as "elasticity_matrix.pdf".
```

7 Numerical Perturbation Check

Use a small perturbation to test the sensitivity approximation.

Choose one nonzero matrix entry a_{ij} and let

$$L_\varepsilon = L + \varepsilon E_{ij},$$

where E_{ij} has a 1 in position (i, j) and zeros elsewhere.

```
i = 0  
j = 3  
epsilon = 1e-5  
  
L_perturbed = L.copy()  
L_perturbed[i, j] += epsilon  
  
perturbed_eigvals = eig(L_perturbed)[0]  
  
# TODO:  
# Compute the perturbed dominant eigenvalue.  
lambda_perturbed =  
  
observed_change = lambda_perturbed - lambda1  
predicted_change = sensitivity[i, j] * epsilon  
  
print("Observed change:", observed_change)  
print("Predicted change:", predicted_change)
```

Questions

1. How closely does the first-order prediction agree with the observed change?
2. What happens if ε is made larger?
3. Why is sensitivity described as a local measure?

Interpretation

Write a short report addressing the following questions.

1. What patterns do you observe in the sensitivity heatmap? Which entries have the largest absolute influence on the dominant eigenvalue?

2. What patterns do you observe in the elasticity heatmap? How do they differ from the sensitivity heatmap?
3. Which age classes appear to have the greatest influence on long-term population growth? Explain your reasoning.

Deliverables

1. A completed Jupyter notebook;
2. The dominant eigenvalue and normalized left and right eigenvectors;
3. The numerical sensitivity and elasticity matrices;
4. Annotated sensitivity and elasticity heatmaps;
5. A numerical perturbation check;
6. A one-page report with an interpretation of the results.

Possible extension

Choose one of the following.

1. Compare the first-order sensitivity prediction with actual changes in λ for perturbations of 1%, 5%, and 10%.
2. Rank all biologically meaningful entries by sensitivity and elasticity and explain differences in the rankings.